

Car Dealership Database

A Relational Database Design and Query Analysis Project

Owais Jawaid

Microsoft SQL Server

May 2023

Database Design:

Database System Objectives -

The Database will show the inventory of new and used cars. The objective of the database is to help customers find the exact car they are looking for. The database would narrow the search results based on features and categories.

Database would be able to store data for the used cars customers would be selling at the dealership based on their make, model, and year. Database would also be storing the past history of the used vehicles, for instance if the vehicle was ever involved in any accident or if it was ever stolen etc.

Database would be able to maintain the data for customers who will bring their vehicles for regular vehicle service or maintenance. For instance, it will include the oil change, tire replacements, or any mechanical issues etc.

Database will store the details of the service date/time, type of service, time required for service, etc. It will also store data for car cleaning services, for example, car wash or car vacuum service etc.

User Requirements -

The user will be able to search for cars based on preference. For instance, a user might want a car to have leather seats of their color of choice, a built-in GPS, or phone connectivity features such as Apple CarPlay, Android Auto, and SiriusXM, etc., then he or she could search for those features to get desired results. Customers would also be able to search for cars based on car's make, model, year, and mileage. The database would be able to sort/filter the search results according to the customer's requirements.

Business Rules -

One car can have one category.

One category can have many cars.

One car can have many services.

One service can be given to one car at a time.

One customer can purchase many cars.

One car can be purchased by one customer.

One car can have many features.

One feature can be in one car.

One customer can have many maintenances.

One maintenance can be performed for one car at a time.

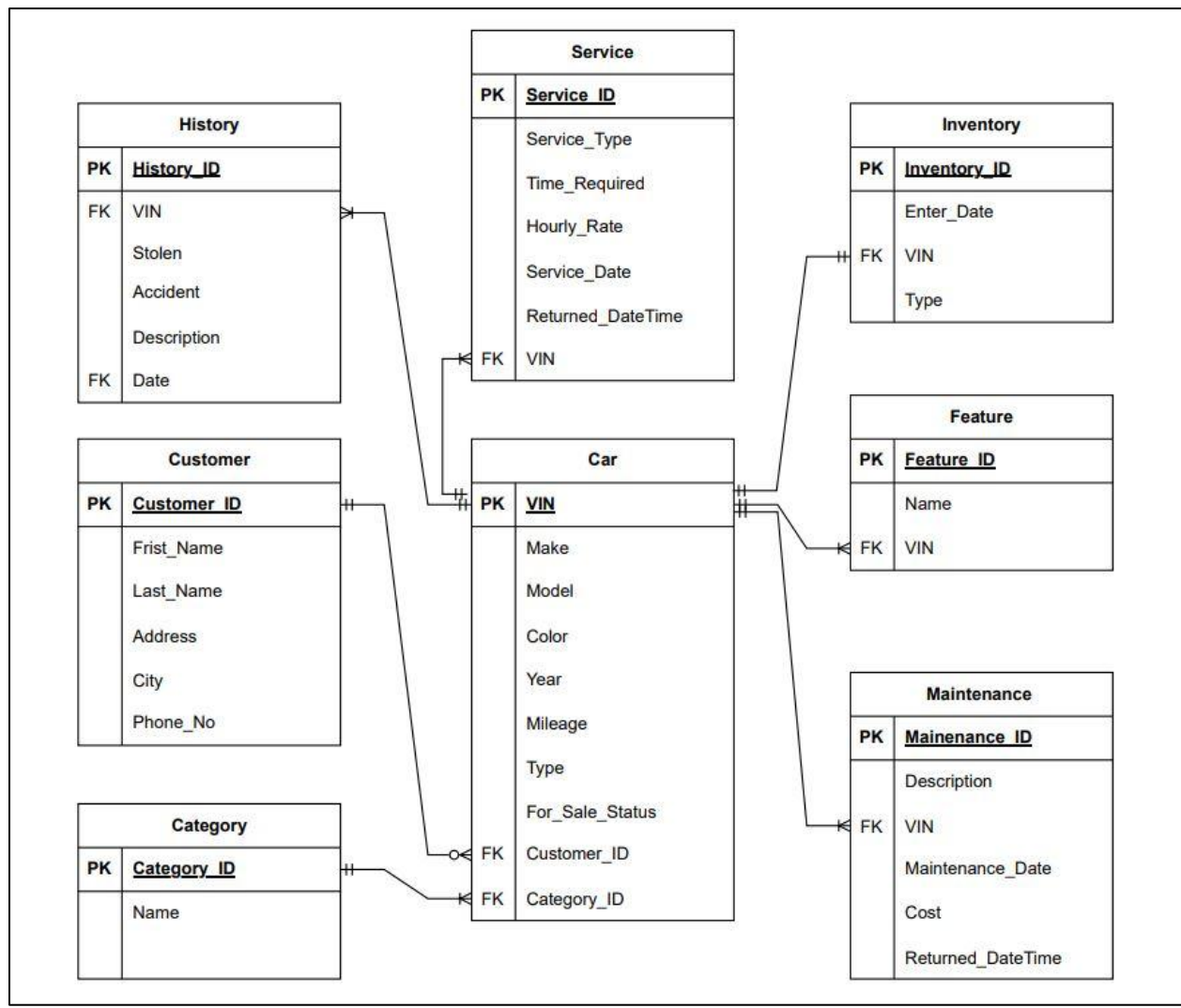
One car with a VIN number can be in one inventory.

One inventory can have one car with one VIN number.

One car can have many histories.

One history can be provided for one car.

Entity-Relationship Diagram (ERD) -



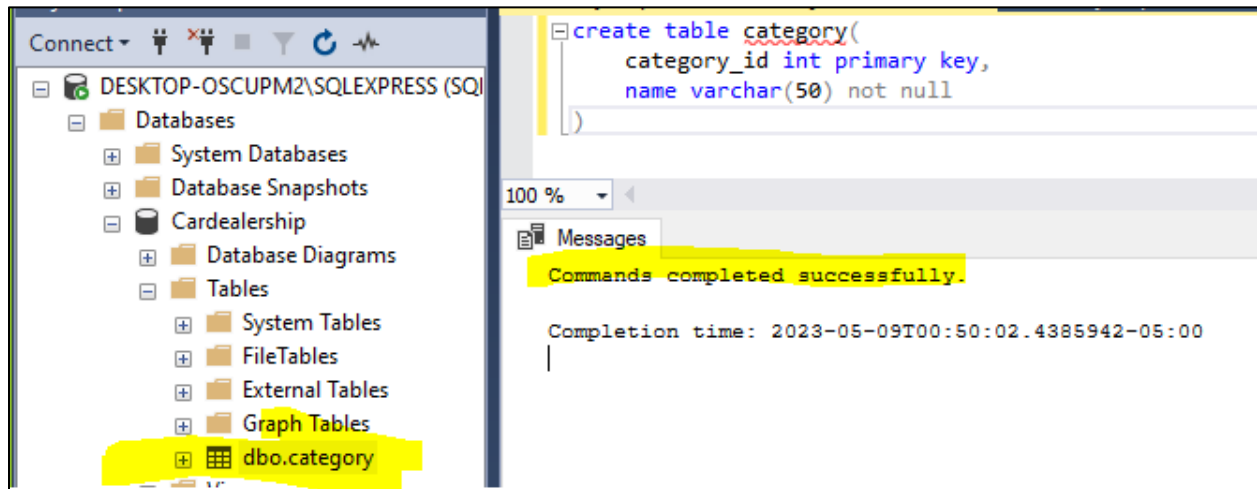
System Data Dictionary -

CarDealership						
TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK OR FK	FK REFERENCED TABLE	REQUIRED
customer	customer_id	Customer id	Int	PK		Y
	first_name	customer first name	varchar(50)			Y
	last_name	customer last name	varchar(50)			
	address	customer address	varchar(50)			
	phone_no	customer phone no	varchar(20)			
	city	customer city	varchar(50)			
Car	VIN	car VIN number	varchar(15)	PK		
	make	car make	varchar(20)			Y
	model	car model	varchar(20)			Y
	color	car color	varchar(20)			
	year	car year	int			Y
	mileage	car mileage	int			
	type	car type	varchar(20)			Y
	for_sale_status	car sale status	varchar(50)			Y
	customer_id	customer customer id	int	FK	customer	Y
History	category_id	customer category id	int	FK	category	Y
	history_id	History id	int	PK		
	stolen	car stolen	int			
	accident	car accident	int			
	description	incident description	varchar(50)			
	date	incident date	date			
feature	VIN	car VIN number	varchar(15)	FK	car	Y
	feature_id	Feature id	int	PK		Y
	name	Feature name	varchar(50)			Y
	VIN	car VIN number	varchar(15)	FK	car	Y
inventory	inventory_id	Inventory id	int	PK		Y
	type	car type	varchar(50)			Y
	enter_date	car enter date	date			Y
	VIN	car VIN number	varchar(15)	FK	car	Y
maintenance	maintenance_id	Maintenance id	int	PK		Y
	description	Maintenance description	varchar(50)			Y
	VIN	car VIN number	varchar(15)	FK	car	Y
	maintenance_date	car maintenance date	date			Y
	cost	maintenance cost	int			Y
	returned_datetime	car return date and time	datetime			Y
service	service_id	Service id	int	PK		Y
	service_type	car service type	varchar(50)			Y
	time_required	service time required	varchar(20)			
	VIN	VIN number	varchar(15)	FK	car	Y
	service_date	car service date	date			Y
	Hourly_rate	service hourly rate	int			Y
Category	returned_datetime	car return date	datetime			Y
	category_id	Category id	int	PK		Y
	name	category name	varchar(50)			Y

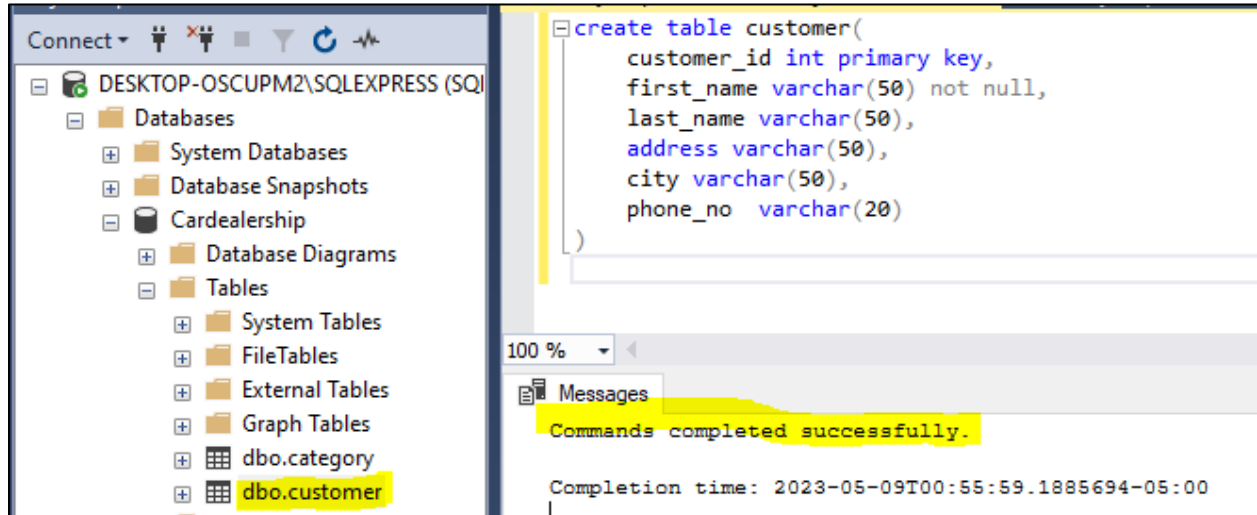
Database System:

SQL Create Table Statements –

Statement to create Category table:



Statement to create customer table:



Statement to create car table:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under the 'Cardealership' database is expanded, and 'dbo.car' is highlighted. The right pane displays the SQL script for creating the 'car' table. The script defines columns: VIN (varchar(15), primary key), make (varchar(20), not null), model (varchar(20), not null), color (varchar(20)), year (int, not null), mileage (int), type (varchar(20), not null), for_sale_status (int, not null), customer_id (int), and category_id (int, not null). It also includes two foreign key constraints: one for customer_id referencing customer(customer_id) and one for category_id referencing category(category_id). The Messages pane at the bottom shows 'Commands completed successfully.' and the completion time: 2023-05-09T00:58:08.2568122-05:00.

```
create table car(  
    VIN varchar(15) primary key,  
    make varchar(20) not null,  
    model varchar(20) not null,  
    color varchar(20),  
    year int not null,  
    mileage int,  
    type varchar(20) not null,  
    for_sale_status int not null,  
    customer_id int,  
    category_id int not null,  
    foreign key (customer_id) references customer(customer_id),  
    foreign key (category_id) references category(category_id)  
)
```

Commands completed successfully.

Completion time: 2023-05-09T00:58:08.2568122-05:00

Statement to create history table:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under the 'Cardealership' database is expanded, and 'dbo.history' is highlighted. The right pane displays the SQL script for creating the 'history' table. The script defines columns: history_id (int, primary key), Stolen (int), Accident (int), Description (varchar(50)), date (date), VIN (varchar(15)), and a foreign key constraint for VIN referencing car(VIN). The Messages pane at the bottom shows 'Commands completed successfully.' and the completion time: 2023-05-09T00:59:47.0820165-05:00.

```
create table history(  
    history_id int primary key,  
    Stolen int,  
    Accident int,  
    Description varchar(50),  
    date date,  
    VIN varchar(15),  
    foreign key (VIN) references car(VIN)  
)
```

Commands completed successfully.

Completion time: 2023-05-09T00:59:47.0820165-05:00

Statement to create feature table:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under the 'Cardealership' database is expanded, listing 'dbo.car', 'dbo.category', 'dbo.customer', 'dbo.feature', and 'dbo.history'. The 'dbo.feature' table is highlighted. On the right, the SQL query editor displays the following T-SQL statement:

```
create table feature(  
    feature_id int primary key,  
    name varchar(50) not null,  
    VIN varchar(15) not null,  
    foreign key (VIN) references car(VIN)  
)
```

Below the query editor, the 'Messages' pane shows the following output:

```
Commands completed successfully.  
  
Completion time: 2023-05-09T01:01:12.7931084-05:00
```

Statement to create inventory table:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under the 'Cardealership' database is expanded, listing 'dbo.car', 'dbo.category', 'dbo.customer', 'dbo.feature', 'dbo.history', and 'dbo.inventory'. The 'dbo.inventory' table is highlighted. On the right, the SQL query editor displays the following T-SQL statement:

```
create table inventory(  
    inventory_id int primary key,  
    type varchar(50) not null,  
    enter_date date not null,  
    VIN varchar(15) not null,  
    foreign key (VIN) references car(VIN)  
)
```

Below the query editor, the 'Messages' pane shows the following output:

```
Commands completed successfully.  
  
Completion time: 2023-05-09T01:02:22.4433208-05:00
```

Statement to create Maintenance table:

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under the 'Cardealership' database is expanded, and 'dbo.Maintenance' is highlighted. The right pane shows the SQL script for creating the table:

```
create table Maintenance(  
    Maintenance_id int primary key,  
    Description varchar(50) not null,  
    VIN varchar(15) not null,  
    Maintenance_Date date not null,  
    Cost int not null,  
    Returned_DateTime datetime not null,  
    foreign key (VIN) references car(VIN)  
)
```

Below the script, the 'Messages' pane shows the execution result:

```
Commands completed successfully.  
  
Completion time: 2023-05-09T01:03:23.6718311-05:00
```

Statement to create service table:

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under the 'Cardealership' database is expanded, and 'dbo.Service' is highlighted. The right pane shows the SQL script for creating the table:

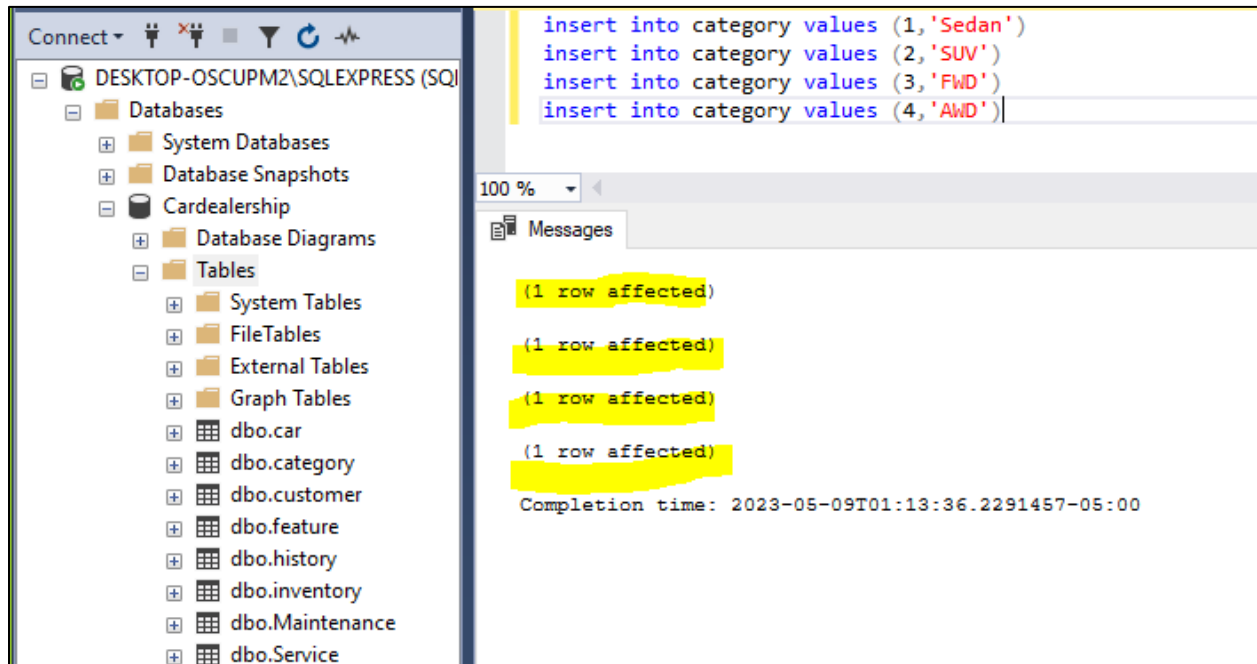
```
create table Service(  
    Service_id int primary key,  
    Service_Type varchar(50) not null,  
    Time_Required varchar(20),  
    VIN varchar(15) not null,  
    Service_Date date not null,  
    Hourly_Rate int not null,  
    Returned_DateTime datetime not null,  
    foreign key (VIN) references car(VIN)  
)
```

Below the script, the 'Messages' pane shows the execution result:

```
Commands completed successfully.  
  
Completion time: 2023-05-09T01:04:26.3517519-05:00
```

Insert Statements:

Statement to insert rows in category table:

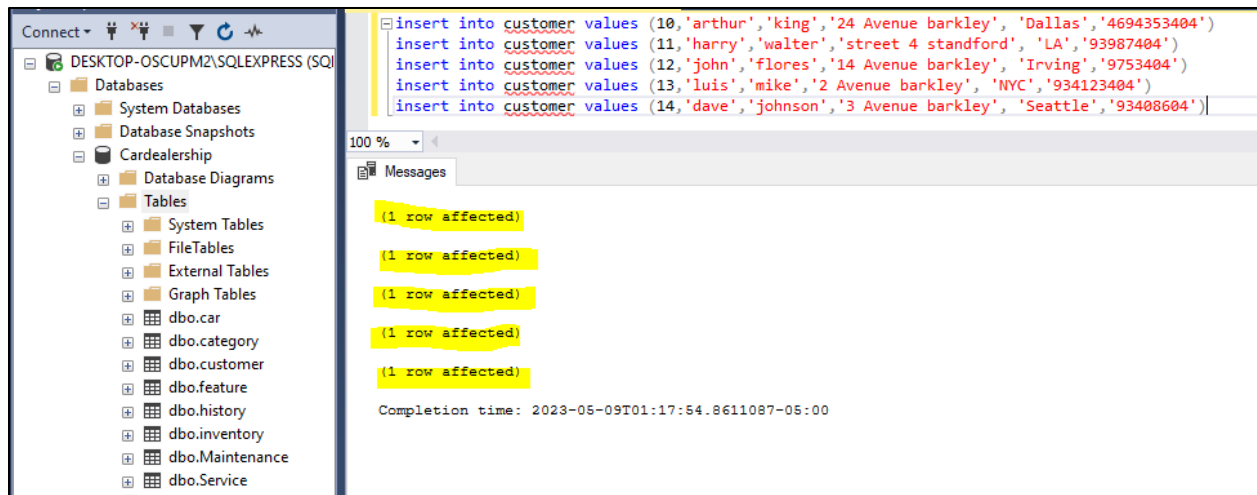


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under the 'Cardealership' database is expanded, listing tables such as dbo.car, dbo.category, dbo.customer, etc. The right pane displays the following SQL statements:

```
insert into category values (1,'Sedan')
insert into category values (2,'SUV')
insert into category values (3,'FWD')
insert into category values (4,'AWD')
```

Below the statements, the 'Messages' pane shows four separate messages, each indicating '(1 row affected)'. The completion time is 2023-05-09T01:13:36.2291457-05:00.

Statement to insert rows in customer table:



The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under the 'Cardealership' database is expanded. The right pane displays the following SQL statements:

```
insert into customer values (10,'arthur','king','24 Avenue barkley', 'Dallas','4694353404')
insert into customer values (11,'harry','walter','street 4 standford', 'LA','93987404')
insert into customer values (12,'john','flores','14 Avenue barkley', 'Irving','9753404')
insert into customer values (13,'luis','mike','2 Avenue barkley', 'NYC','934123404')
insert into customer values (14,'dave','johnson','3 Avenue barkley', 'Seattle','93408604')
```

Below the statements, the 'Messages' pane shows five separate messages, each indicating '(1 row affected)'. The completion time is 2023-05-09T01:17:54.8611087-05:00.

Statement to insert rows in car table:

```
insert into car values ('AAA342J1','Honda','City Aspire','Black',2023,14,'Used',1,10,1)
insert into car values ('FDS342J2','Toyota','Corolla GLI','White',2020,12,'Used',1,11,1)
insert into car values ('VSD342J3','Audi','A6','Red',2023,14,'New',1,12,1)
insert into car values ('ASXA342J','Mercedes','Benz','Silver',2023,14,'New',1,13,1)
insert into car values ('VDF342J4','Toyota','Fortuner Sigma','Black',2022,9,'Used',1,14,4)
```

100 %

Messages

(1 row affected)

(1 row affected)

(1 row affected)

(1 row affected)

(1 row affected)

Completion time: 2023-05-09T01:20:57.1187592-05:00

Statement to insert rows in feature table:

```
insert into feature values (1,'Apple Car Play','VDF342J4')
insert into feature values (2,'Android Auto','VDF342J4')
insert into feature values (3,'GPS','VDF342J4')
insert into feature values (4,'Bluetooth','VDF342J4')
insert into feature values (5,'GPS','ASXA342J')
```

100 %

Messages

(1 row affected)

(1 row affected)

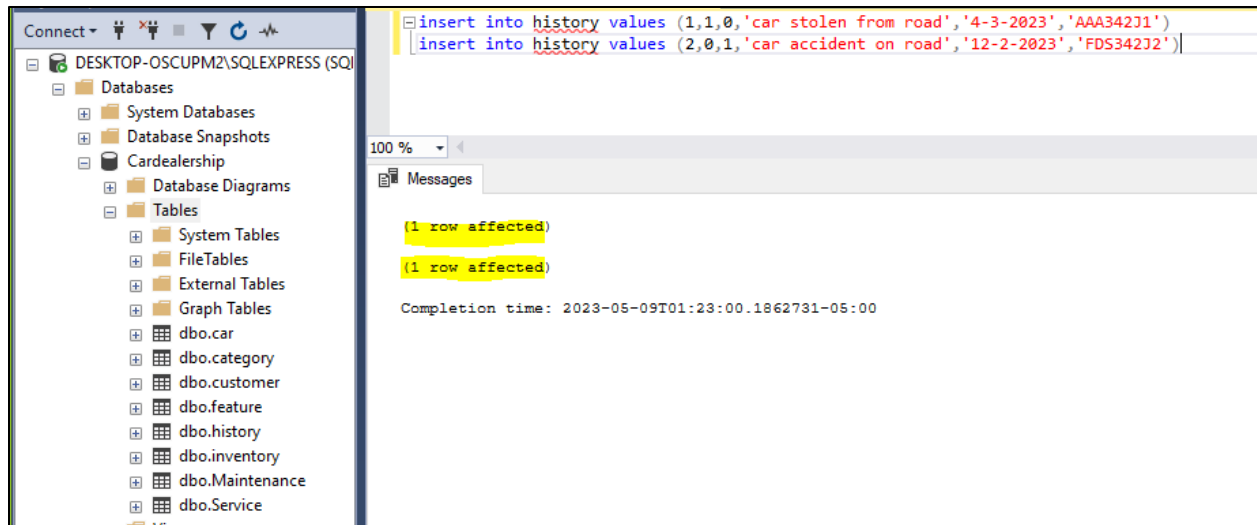
(1 row affected)

(1 row affected)

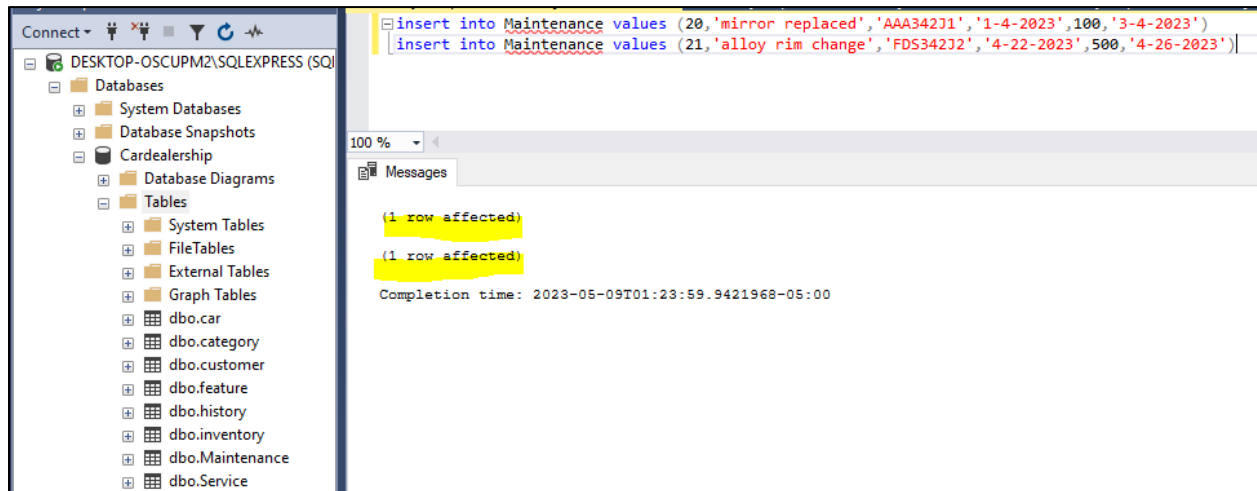
(1 row affected)

Completion time: 2023-05-09T01:21:57.9800758-05:00

Statement to insert rows in history table:



Statement to insert rows in Maintenance table:



Statement to insert rows in inventory table:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under 'Cardealership' is expanded, showing tables like dbo.car, dbo.category, and dbo.inventory. The right pane displays the following SQL statements:

```
insert into inventory values (11,'Not Sold','12-3-2023','AAA342J1')
insert into inventory values (12,'Not Sold','12-3-2023','FDS342J2')
insert into inventory values (13,'Not Sold','12-3-2023','AAA342J1')
insert into inventory values (14,'Not Sold','12-3-2023','AAA342J1')
insert into inventory values (15,'Not Sold','12-3-2023','AAA342J1')
```

The Messages pane shows the following output:

```
(1 row affected)
(1 row affected)
(1 row affected)
(1 row affected)
(1 row affected)
```

Completion time: 2023-05-09T01:24:55.1233728-05:00

Statement to insert rows in service table:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Tables' folder under 'Cardealership' is expanded, showing tables like dbo.car, dbo.category, and dbo.service. The right pane displays the following SQL statements:

```
insert into service values (30,'Car Wash','1 hour','FDS342J2','4-22-2023',10,'4-22-2023 15:23:11')
insert into service values (31,'Car vacuum','1 hour','AAA342J1','4-4-2023',12,'4-4-2023 11:23:11')
insert into service values (32,'Car Wash','1 hour','FDS342J2','3-4-2023',10,'3-4-2023 15:23:11')
insert into service values (33,'Car Wash','1 hour','FDS342J2','2-4-2023',10,'2-4-2023 19:23:11')
```

The Messages pane shows the following output:

```
(1 row affected)
(1 row affected)
(1 row affected)
(1 row affected)
```

Completion time: 2023-05-09T01:25:53.4084305-05:00

SQL Select Statements –

Query retrieves all customers from the city of Dallas:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded, showing the 'CarDealership' database. The 'Tables' folder is also expanded, listing various tables including 'dbo.customer'. The main query window displays the following SQL statement:

```
Select * from customer where city = 'Dallas'
```

The 'Results' tab shows the output of the query as a table with the following data:

	customer_id	first_name	last_name	address	city	phone_no
1	10	arthur	king	24 Avenue barkley	Dallas	4694353404

Query displays all cars which have a GPS feature:

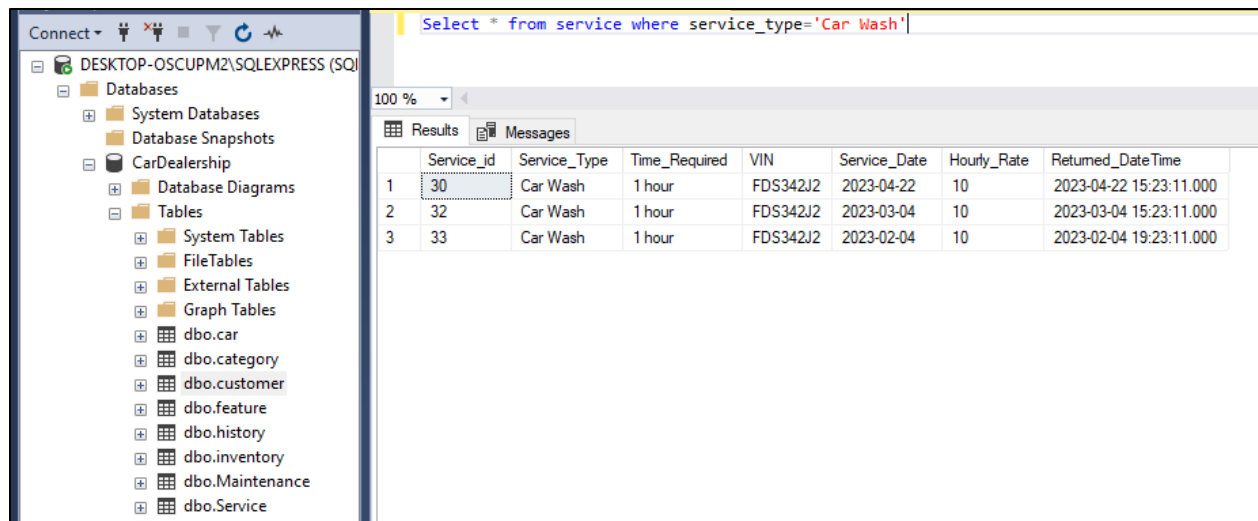
The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded, showing the 'CarDealership' database. The 'Tables' folder is also expanded, listing various tables including 'dbo.feature'. The main query window displays the following SQL statement:

```
Select * from feature where name= 'GPS'
```

The 'Results' tab shows the output of the query as a table with the following data:

	feature_id	name	VIN
1	3	GPS	VDF342J4
2	5	GPS	ASXA342J

Query shows list of cars that use the car wash service:



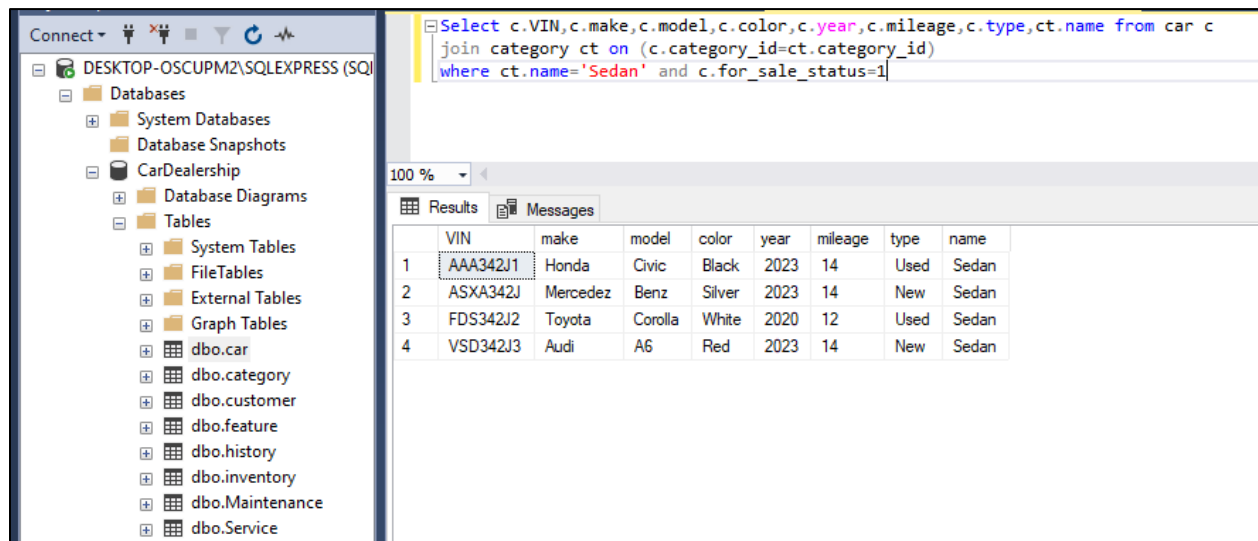
The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the database structure for 'DESKTOP-OSCUPM2\SQLEXPRESS (SQL)' with the 'CarDealership' database selected. The right pane shows a query window with the following SQL statement:

```
Select * from service where service_type='Car Wash'
```

The query results are displayed in a table with the following columns: Service_id, Service_Type, Time_Required, VIN, Service_Date, Hourly_Rate, and Returned_DateTime. The results show three records for car wash services.

	Service_id	Service_Type	Time_Required	VIN	Service_Date	Hourly_Rate	Returned_DateTime
1	30	Car Wash	1 hour	FDS342J2	2023-04-22	10	2023-04-22 15:23:11.000
2	32	Car Wash	1 hour	FDS342J2	2023-03-04	10	2023-03-04 15:23:11.000
3	33	Car Wash	1 hour	FDS342J2	2023-02-04	10	2023-02-04 19:23:11.000

Query pulls list of sedan cars that are ready for sale:



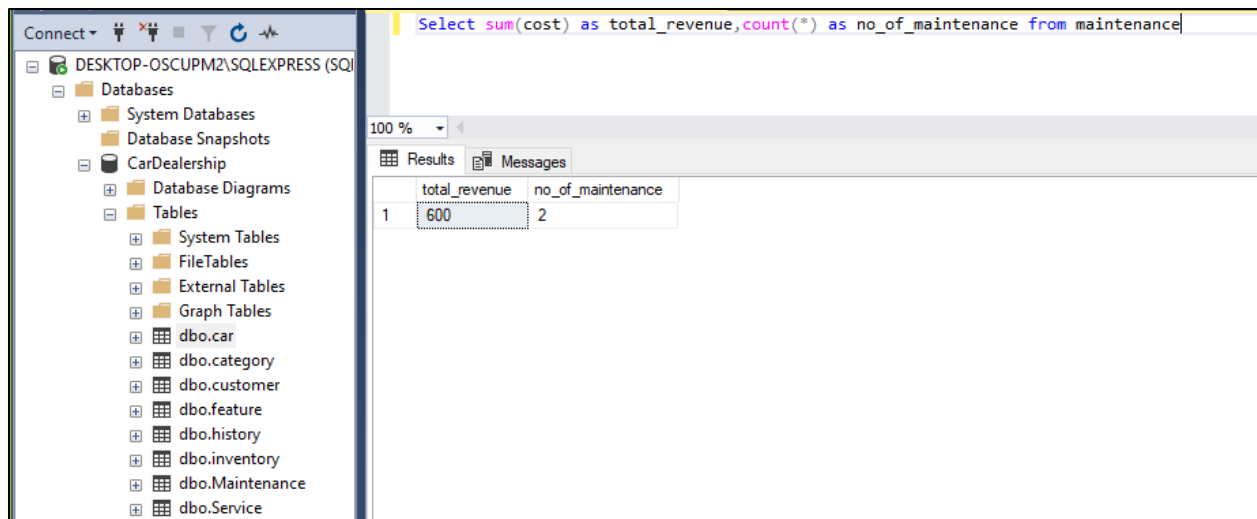
The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the database structure for 'DESKTOP-OSCUPM2\SQLEXPRESS (SQL)' with the 'CarDealership' database selected. The right pane shows a query window with the following SQL statement:

```
Select c.VIN,c.make,c.model,c.color,c.year,c.mileage,c.type,ct.name from car c  
join category ct on (c.category_id=ct.category_id)  
where ct.name='Sedan' and c.for_sale_status=1
```

The query results are displayed in a table with the following columns: VIN, make, model, color, year, mileage, type, and name. The results show four records for sedan cars that are ready for sale.

	VIN	make	model	color	year	mileage	type	name
1	AAA342J1	Honda	Civic	Black	2023	14	Used	Sedan
2	ASXA342J	Mercedes	Benz	Silver	2023	14	New	Sedan
3	FDS342J2	Toyota	Corolla	White	2020	12	Used	Sedan
4	VSD342J3	Audi	A6	Red	2023	14	New	Sedan

Query will show total revenue generated from car maintenance:



The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded, showing the 'CarDealership' database and its tables. The 'dbo.Maintenance' table is selected. The central pane shows a SQL query: `Select sum(cost) as total_revenue, count(*) as no_of_maintenance from maintenance`. The 'Results' tab is active, displaying a single row of data: `total_revenue` is 600 and `no_of_maintenance` is 2.

	total_revenue	no_of_maintenance
1	600	2